

Technologies of Interest to SAWG: aligning with a global donor's (USAID's) perspective

Rahul Bhargava

Member, South Asia Working Group
IEEE Smart Village

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- SDGs and Integrated development

- Integrated approaches

USAID Powering Agriculture 2012–19

USAID/Nepal KISAN II

Nepal KISAN II Activities

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- Pay-Per-Use Irrigation Service with Solar Pump

- BioLite HomeStove

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- Custom Hiring Centres: EM3 AgriServices

- Zero tillage wheat

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SDGs and Integrated development

The SDGs provide an **integrated** and **multidisciplinary** policy framework whose implementation requires whole-of-government and whole-of-society approaches.¹

Institutions and leaders at all levels of government and public administrations need to focus on

- people,
- innovation,
- partnerships and
- data,

to shift from traditional policy and delivery tools to those more conducive to experimentation and citizens' feedback.

¹UN, Capacity Development Programme Management Office, Department of Economic & Social Affairs.

Integrated approaches

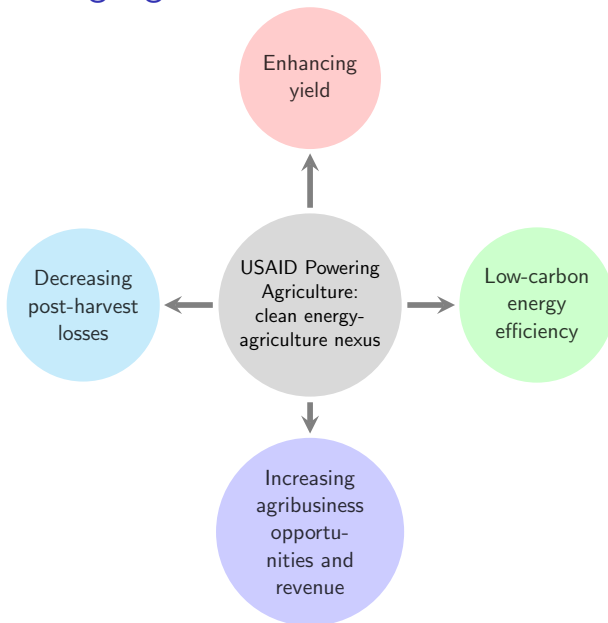
Development initiatives need to be more holistic – the approaches may need to be as interconnected as the problems.²

Integrated approaches with promising evidence include:

- linking economic strengthening and health efforts;
- incorporating water, sanitation and hygiene (WASH),
- health or nutritional support into schools;
- combining nutrition and agriculture initiatives; and
- integrating family planning with environmental programming.

²Tricia Petruney, FHI 360.

USAID Powering Agriculture 2012–19



USAID/Nepal KISAN II

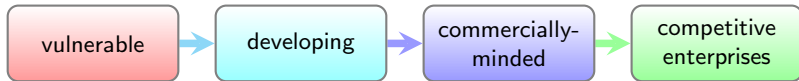
KISAN II employs a “push-pull” approach to engagement.

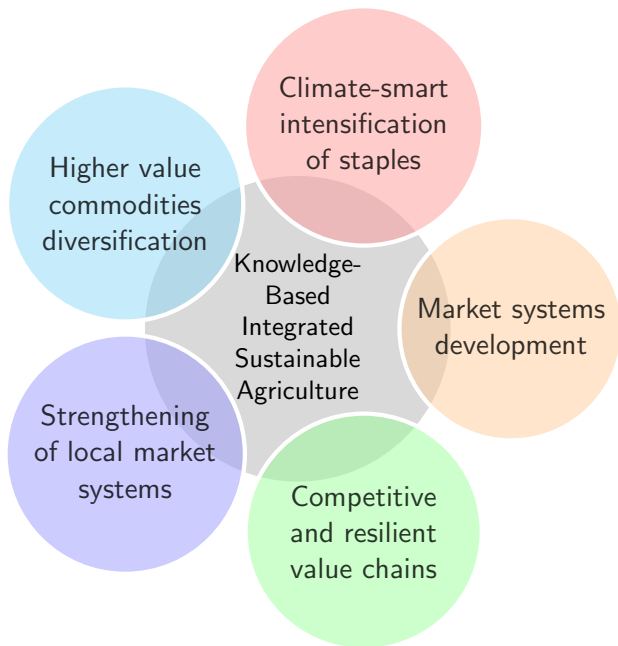
Push to help poor farmers, individuals build capacity through

- intensification, diversification, and
- *value addition* activities

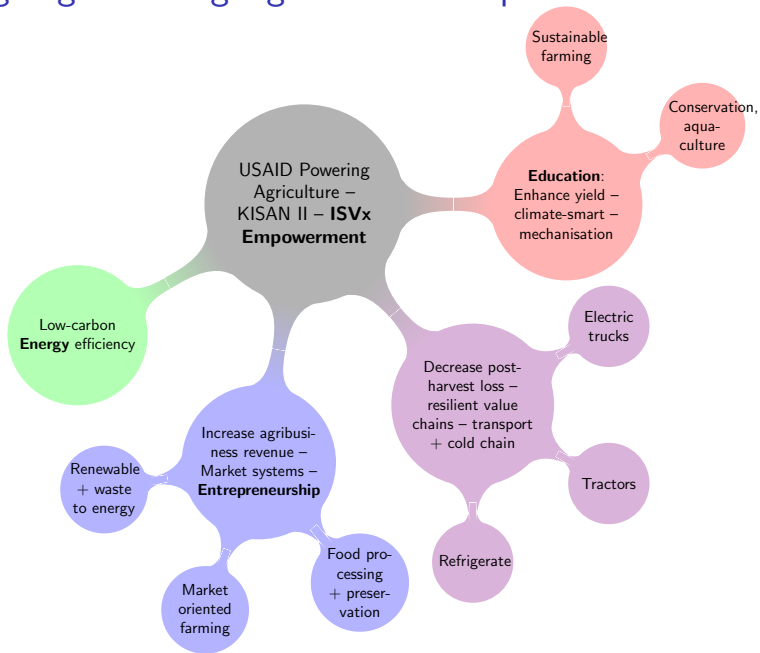
Pull to increase demand for smallholder production, labour, related goods & services; improve affordability, accessibility of skills, resources, inputs, and *supporting services*.

Approach to **empower** and graduate farming households into agricultural enterprises evolving from





Aligning Powering Agriculture – Nepal KISAN II – ISVx



Nepal KISAN II Activities

1. Increase adoption of profitable, productivity-enhancing, and climate smart technologies.
2. Strengthen firms and SMEs to support selected market systems.
3. Enhance financial services markets and infrastructure.
4. Build capacity to support market systems development.
5. Enhance literacy and business development skills.

Low-cost solar pumps for field crops

Focus Areas Agriculture, Cultivation & Tools, Seeds & Stock, Soil Management and Agriculture Water Management

Implemented in India, Nepal and Bangladesh

Problem High energy costs for pumping groundwater for irrigation. Without government subsidies farmers are not willing to purchase solar pumps.

Funder USAID

Solution CSISA³ rapidly developed and tested 1 kW Sasto Solar Sichai (cheaper solar irrigation) pumps for the irrigation of field crops like wheat, maize and rice. Using off-the-shelf 48V DC motors, charge controllers used in transport Nepal (battery-powered three-wheelers), 2-3" centrifugal pumps, and 1200W solar PV panels, the cost brought down from US\$ 2500 to less than US\$ 1500.

³<https://csisa.org/>

Pay-Per-Use Irrigation Service with Solar Pump

Focus Areas Agriculture, Energy and Solar

Implemented in India

Problem Smallholders access to solar irrigation.

Funder USAID, Duke Energy, SIDA, BMZ⁴ and OPIC⁵

Solution Claro Energy will offer a pay-per-use irrigation service for portable solar pumps.⁶ For affordable, convenient, and on-demand irrigation. For farmers without pumps, without upfront costs who will call toll-free, pre-pay, and schedule irrigation on their fields. In India, irrigation water is dependent on the monsoon or electricity to operate ground water pumps. Villages lack access to electricity. Diesel pumps are often used but have drawbacks, including environmental pollution, fuel costs.

⁴Federal Ministry for Economic Cooperation and Development

⁵Overseas Private Investment Corporation

⁶<http://poweringag.org/innovators/>

BioLite HomeStove I

Focus Areas Post Harvest, Economic Empowerment, Lighting, Energy, Biomass/BioFuel, Distributed Generation Systems and Energy Efficiency

Implemented in India, Uganda, Rwanda, Tanzania and Kenya

Problem 3 billion cook meals over open biomass fires. Toxic smoke leads to over 4 million respiratory-related deaths every year and releases significant amounts of CO₂, producing 25% of global black carbon emissions. Households relying on open fire cooking lack reliable access to electricity; 1.2 billion people have no access to electricity. Forced to use dirty, expensive kerosene lamps for lighting.

Funder USAID

BioLite HomeStove II

Solution A fan-assisted biomass stove that burns wood like gas, turns fire into electricity, and reduces smoke emissions by 90% and fuel consumption by 50% compared to traditional open-fire cooking. The HomeStove co-generates 2W of electricity from the flame using thermoelectric generation, enabling users to charge mobile phones and lights via a USB port. The HomeStove is also compatible with locally available fuels (e.g. wood, cow dung, or crop waste), and consistent with local cooking practices (side-fed operation). Rocket stoves (Envirofit, BURN, EcoZoom), at best achieve 50% emissions reductions—lessening their climate and health impact—while fan stoves (BP, Philips) require external electricity for operation, specialised fuel pellets, and

BioLite HomeStove III

unfamiliar top-fed operation. LPG stoves, though clean burning, are 10–20x more expensive than the HomeStove over three years and have poor penetration among low-income households. In comparison, the BioLite HomeStove is the only ultra-clean, biomass stove that can co-generate electricity from the cooking flame to power LED lights and mobile phones. The HomeStove features a design that is both aspirational and durable, while remaining intuitive for BoP users without requiring significant behaviour change to operate.

Hybrid Vehicles w/Exportable Power for Mechanisation I

Focus Areas Agriculture, Energy, Cultivation & Tools, Processing & Production, Energy Efficiency, Solar, Wind & Water Energy and Technology

Implemented in India

Problem Although farm productivity in India has grown steadily over recent decades, crop yields are still just 30–60% of the best crop yields in many other countries. Agricultural productivity could be significantly boosted by creating more accessible agricultural machinery, and by providing a mobile energy source that runs on inexpensive renewables to meet multiple farm needs such as harvesting, cold storage, and transportation.

Funder USAID, Duke Energy, SIDA⁷, BMZ and OPIC

Hybrid Vehicles w/Exportable Power for Mechanisation II

Solution Motivo⁸ is developing a “Swiss-Army Knife” system—the Hybrid Agriculture/Road Vehicles with Electricity Storage and Transformation (HARVEST)—to solve a range of agricultural mechanisation and power-related problems. HARVEST is a multi-purpose platform that provides power for plowing, well-drilling, cold storage, and transporting crops to market. The system utilises power from varied energy sources such as solar panels, wind turbines, micro-hydro turbines, or the grid to enable increased productivity all along the agriculture value chain. The entire system is operated at the community level, and facilitated by mobile communication technology for scheduling, billing, and payments.

⁷Swedish International Development Cooperation Agency

⁸<http://www.motivoengineering.com/>

Custom Hiring Centres: EM3 AgriServices

Focus Areas Agriculture, Cultivation & Tools

Implemented in India

Problem Smallholders access to farm machinery.

Funder Global Innovation Fund

Solution Pay-per-use farm mechanisation services company.
An innovative business model that formalises existing informal channels, EM3 make farming mechanisation affordable for the first time for millions of rural poor Indian farmers. EM3 connects equipment owners with smallholder farmers, offering on-demand, pay-per-use services of harvesters, tractors and farming equipment. Through EM3's platform, smallholder farmers can access services for land preparation, sowing and harvesting, billed per acre or per hour.

Zero tillage wheat

Focus Areas Agriculture, Energy and Solar

Implemented in India, Bangladesh and Nepal

Problem High costs of cultivation and inefficient resource use.

Funder U.S. Agency for International Development

Solution Across South Asia and as a first entry point for 'conservation agriculture' based crop establishment, CSISA⁹ has worked to refine, evaluate, and scale zero tillage technologies for several staple crops, with special attention given to wheat. Research results consistently demonstrate that zero-till wheat yields about 0.5 tons per hectare more while reducing production costs by up to 45% when implemented through service providers. In Nepal, CSISA is working with machinery importers and service providers to increase access to this proven technology.

⁹CSISA <http://www.csisa.org/>

Mechanical transplanting for rice

Focus Areas Cultivation & Tools

Implemented in India, Bangladesh and Nepal

Problem Labour scarcity at the time of rice establishment.

Funder USAID

Solution Substitutes mechanical technology for labour. Labor or capital constrained rice farmers are connected to mechanisation value chains through service providers. Less risky than directly-sown rice. CSISA is led by the International Maize and Wheat Improvement Center (CIMMYT) and implemented by a team of scientists and change-makers with the International Food Policy Research Institute (IFPRI) and the International Rice Research Institute (IRRI). Operating across South Asia, CSISA works to increase the adoption of resource-conserving and climate-resilient technologies and improves farmers' access to markets and enterprise development.

Subsurface Drip Irrigation System

Focus Areas Agriculture, Environment and Water Sanitation and Hygiene (WASH)

Implemented in Ethiopia and India

Problem Efficient irrigation.

Funder USAID, SIDA, Min. of Foreign Affairs of the Netherlands, DST of the Govt. of South Africa

Solution Delivers assured spread moisture at plant root zone to cultivate vegetables, flowers, fruit/forestry trees using only one fifth of water compared to Indian drip irrigation systems. Fosters soil micro-organisms. Automated but no electricity (gravity based), no weed growth/surface salt accumulation, enhances soil nutrients, uses harvested/stored water, reaches irrigation to low rainfall areas transforming livelihoods of poor farmers and grow more food.

NewSil

Focus Areas Agriculture, Environment and Water Sanitation and Hygiene (WASH)

Implemented in South Africa and India

Problem Water use efficiency.

Funder USAID, SIDA, Min. of Foreign Affairs of the Netherlands, DST of the Govt. of South Africa

Solution Si Technologies have found a way to stabilise silicic acid that allows food crops to absorb Silicon, resulting in a reduction of water consumption of 30–50%. The product is called NewSil. Silicon is an important element in plant growth. Only silicic acid $\text{Si}(\text{OH})_4$ can directly be absorbed by plants. NewSil is game changing as it allows uptake of a natural product resulting in more food with less water.

Dry chain management for reducing aflatoxin in maize

Focus Areas Agriculture, Cultivation & Tools, Seeds & Stock, Soil Management and Agriculture Water Management

Implemented in India, Nepal and Bangladesh

Problem Aflatoxin contamination of maize and the associated food chain in Nepal.

Funder USAID

Solution Pre-harvest aflatoxin levels can be reduced with better fertiliser management and post-harvest aflatoxin levels can be reduced by taking maize grain off the cob and sun drying the grain before storage.¹⁰

¹⁰CSISA for the International Maize and Wheat Improvement Center (CIMMYT). <https://www.cimmyt.org/>

Solar Conduction Dryer

Focus Areas Agriculture, Economic Growth and Trade, Energy and Gender

Implemented in Vietnam, Sri Lanka, Kenya and India

Problem Aflatoxin contamination of maize and the associated food chain in Nepal.

Funder Micheal and Susan Dell Foundation

Solution The patented Solar Conduction Dryer cost half that of other solar dryers; the world's most efficient dryer. Electrical dryers: cost required for drying is \$0.2/kg. SCD pays back against electrical dryers in 100–200 days. Food Quality: SCD produces dehydrated products with retention of 80–90% of key nutrients and meets international standards in terms of microbial count and aflatoxin levels. Results are tested and verified by international food laboratories.

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¹¹Science 4 Society Technologies <http://s4stechnologies.com/>

Akashganga – Automatic Milk Collection Systems

Focus Areas Food Safety and Standards and Good Governance

Implemented in Nepal, Vietnam, Uganda, Kenya and India

Problem An average farmer in India owns 1 or 2 milch animals or 1–2 acres of land. Milk would often get spoilt, with producers waiting in long queues and milk collection performed manually.

Funder NASSCOM Foundation

Solution Automation of milk collection including.¹²

- Weighing milk with electronic weighing scale,
- Fat testing using Milko Tester,
- Capturing unique member ID by PC software,
- Printing pay slips, with all this data and the amount to be paid.

¹²Shree Kamdhenu Electronics Pvt. Ltd. <http://www.akashganga.in/>

Reducing Milk Spoilage w/Solar Powered Milk Chilling

Focus Areas Agriculture, Food Safety and Standards, Livestock & Agriculture, Processing & Production and Solar

Implemented in India

Problem In India, \$10 billion of perishable foods go waste annually due to unreliable cold-chains. No reliable electricity for refrigeration near farms.

Funder USAID, Duke Energy, SIDA, BMZ and OPIC

Solution Promethean's refrigeration solution uses a thermal energy battery pack that charges on solar power and/or a few hours of grid electricity. It then provides cold storage around the clock. Dairy processors collect raw milk chill it in a rapid milk cooler, reducing unchilled time by 75%. Promethean will design and deploy refrigeration systems in collaboration with India's largest private dairy and one of India's largest solar installers¹³

¹³Promethean. <http://www.cooelectrica.com/impact/>

Traceability: Better Cotton Initiative

Focus Areas Agriculture, Economic Growth and Trade, Environment, Processing & Production and Natural Resources

Implemented in Senegal, Pakistan, Mozambique and India

Problem Mitigate environmental impact, high water consumption, natural habitat loss, inadequate farmer capacity from growing cotton.

Funder USAID

Solution BCI has developed the first components of a global traceability system and investment model. It is a holistic approach to more sustainable cotton production, based on a voluntary standard for improvement, which can be implemented in a mainstream way on a global scale. An innovative funding and traceability system secures support to farmers while avoiding trading premiums.¹⁴

¹⁴<http://bettercotton.org/>

Solar Grain Dryer

Focus Areas Agriculture and Processing & Production

Implemented in India

Problem Currently, women dry harvested grain in the sun, by tilting and spreading at regular intervals for up to 45 days, taking time from other farming tasks. Open sun drying reduces quality, is less efficient, leads to significant loss of productivity.

Funder Bill and Melinda Gates Foundation

Solution Vaibhav Tidke has developed a mobile solar powered grain dryer to rapidly reduce moisture of grains, extending storage life. A wheel-loaded movable dryer that dries in one day with no operating cost, its mobility allows sharing between farms reducing capital cost. It is low maintenance and locally fabricated. ¹⁵

¹⁵<http://gchh.grandchallenges.org/grant/solar-grain-dryer>

Precision Agriculture for Development

Focus Areas Agriculture, Evidence & Impact, Technology, Environment, Digital Development, Human Centered Design and Resilience

Implemented in India, Kenya, Rwanda, Pakistan, Ethiopia, Bangladesh, Uganda and Zambia

Problem Non availability of actionable information to smallholder farmers in developing countries.

Funder Bill and Melinda Gates Foundation, Private Foundations

Solution PAD helps smallholder farmers improve productivity, increase profitability, and advance environmental sustainability. PAD is pioneering a new model for agricultural extension: delivering to farmers personalised agricultural advice through their mobile phones.¹⁶

¹⁶<https://precisionag.org>

Plantwise

Focus Areas Agriculture

Implemented in Bolivia, Bangladesh, Uganda, Nicaragua, Rwanda, Vietnam, Sierra Leone, Nepal, Kenya, India, etc.

Problem Yearly, an estimated 40% of crops grown worldwide are lost to pests

Funder UKaid, SWISSAID, Ministry of Foreign Affairs of the Netherlands, EuropeAid, Irish Aid, IFAD, ACIAR, Ministry of Rural and Rural Affairs PRC

Solution Working closely with national agricultural advisory services, Plantwise has established a global plant clinic network run by trained plant doctors, for farmers to find practical plant health advice. Plant clinics work just like clinics for human health: farmers visit with samples of their crops, and plant doctors diagnose the problem and make science-based recommendations.¹⁷

¹⁷<https://www.plantwise.org/>

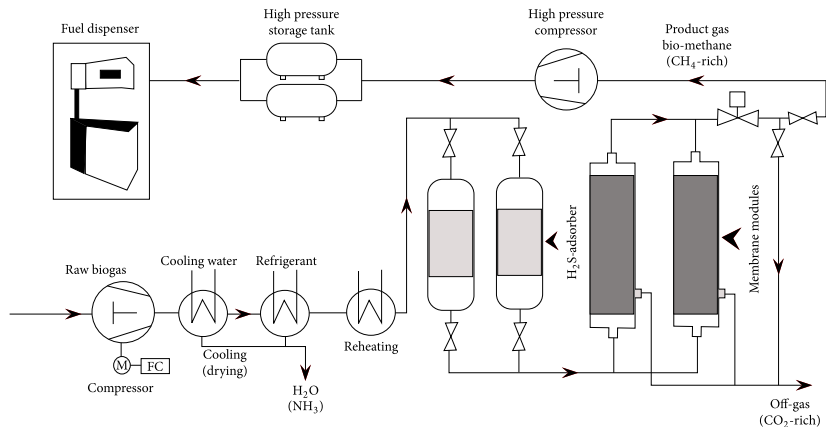


Figure 1: Schematic diagram of the bio-CNG generation process

Transport: UTV

